

Problem Set 6
COMP301 SPRING 2025
Week 6: 24.03.2025- 28.03.2025

Instructions:

- Submit your answers to the LearnHub PS6 assignment until April 5th Saturday, at 23.59.
- Please submit only **one single PDF file**, where all of your written solutions or codes and outputs for each of the parts are included.
- Name your submission file as *id_username_ps6.pdf*

```
Program ::= Expression
         a-program (exp1)

Expression ::= Number
           const-exp (num)

Expression ::= - (Expression , Expression)
           diff-exp (exp1 exp2)

Expression ::= zero? (Expression)
           zero?-exp (exp1)

Expression ::= if Expression then Expression else Expression
           if-exp (exp1 exp2 exp3)

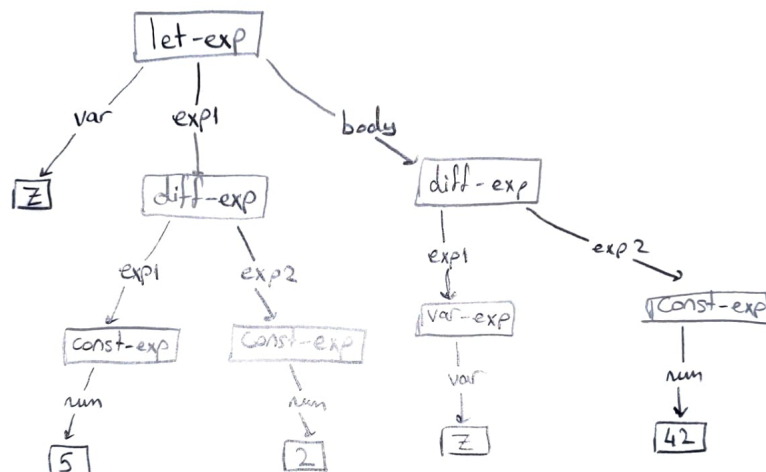
Expression ::= Identifier
           var-exp (var)

Expression ::= let Identifier = Expression in Expression
           let-exp (var exp1 body)
```

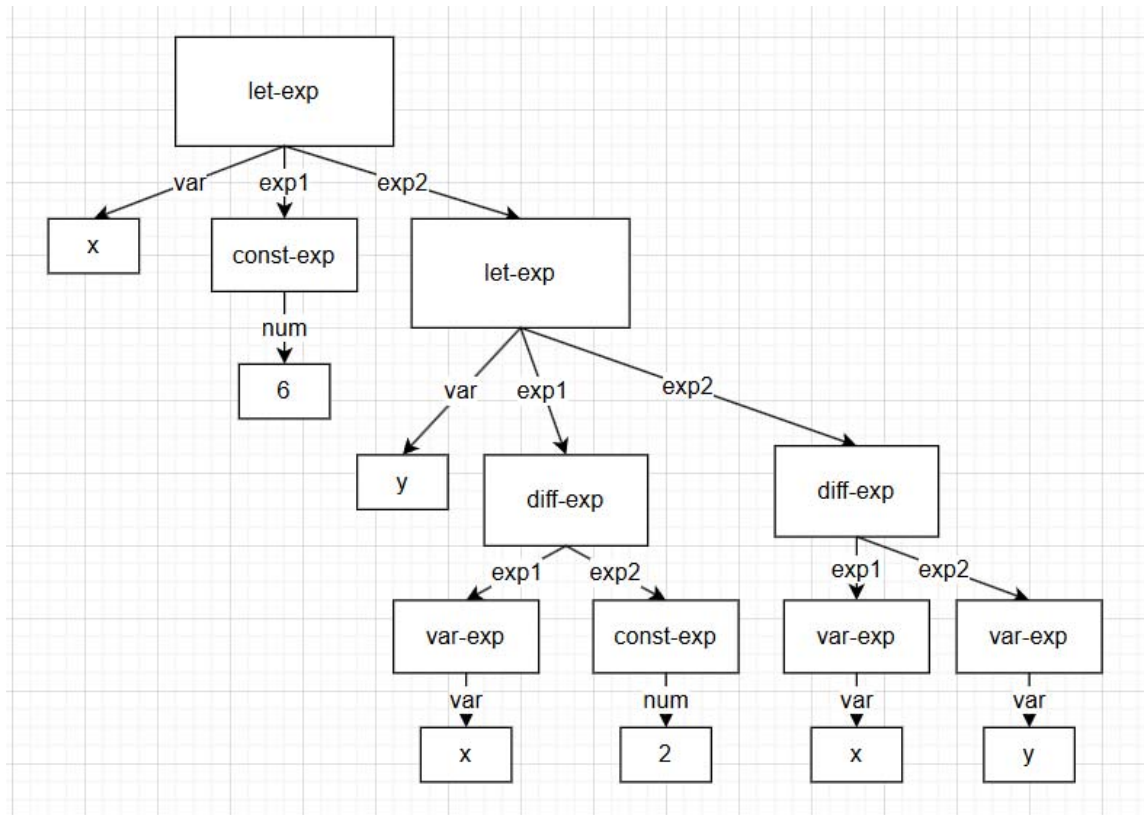
Problem 1: Draw an abstract syntax tree for the given code using the grammar definition above.

```
let x = 6
  in let y = -(x, 2)
    in -(x, y)
```

Problem 2: Write the code that is represented as an abstract syntax tree below.



Question 1



Question 2

```
let z = -(5,2)
  in -(z,42)
```